MIL-O-14559C
13 August 1981
SUPERSEDING
MIL-O-14559B
6 February 1969

MILITARY SPECIFICATION

OIL GUN, PNEUMATIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a hand-operated, pneumatic oil gun designed to produce degrees of atomization from a fine spray or mist to a coarse spray of oil.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

FSC 4930

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DE, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026l0-18T03:26Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-G-1086	- Gasket-Material, Synthetic Rubber (For Bolted Steel Tanks).

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.
FED-STD-H28/7	- Screw-Thread Standards for Federal Services, Section 7, American Standard Pipe Threads (Except Dryseal and Hose Coupling Types).

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The oil gun shall be as shown in Figure 1 (see 6.5), and shall consist of a body and handle assembly, a fluid container, a barrel assembly; and shall be the siphon feed type and atomize the oils specified herein. The weight of the empty oil gun shall be not more than 3.75 pounds. Parts subject to wear, breakage or distortion shall be accessible for adjustment, replacement, and repair.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.3).

Source: https://assist.dla.mil -- Downloaded: 2026l0-18T03:26Z
Check the source to verify that this is the current version before use.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.6).

3.4 Threads. All screw threads shall conform to FED-STD-H28.

3.5 Body and handle assembly. The body and handle shall be die cast zinc or aluminum alloys. The handle shall be the pistol or modified pistol-grip type, designed to allow easy handling and one-hand operation.

3.5.1 Air control valve. The air control valve shall be corrosion-resistant metal and shall be built into the body of the oil gun. The valve shall be actuated by a button or trigger located as to be convenient to the operator's thumb or index finger.

3.5.2 Air adjustment valve. The air adjustment valve shall be corrosion-resistant metal and be located as to be convenient to operate. The valve shall be designed to allow for adjustment of air pressure ranging from line pressure of 50 pounds per square inch gage (psig) to minimum operating pressure.

3.5.3 Container cover. The container cover shall be a flat, recessed surface, built into the underside of the gun body and threaded internally to accommodate the threaded neck of the fluid container. A gasket conforming to MIL-G-1086, Type I shall be provided to insure a leakproof connection.

3.5.4 Air inlet. The air inlet shall be located on the bottom or the back of the oil gun handle or grip. Inlet shall be threaded with a 1/4-18 NPSM pipe thread in accordance with FED-STD-H28/7.

3.6 Fluid container. The fluid container shall have a nominal capacity of 1 quart and shall be steel plated to resist corrosion. The neck of the container shall be large enough to assure easy filling and shall be threaded on the outside. A small vent hole shall be provided in the top of the container or in the container cover portion of the gun body.

3.7 Barrel. The barrel shall be corrosion-resistant steel of rigid construction, not less than 10 inches in length, and shall be secured to the body by a means of soldering, brazing, or a knurled retaining nut.

3.8 Nozzle assembly. The nozzle assembly shall be adjustable to control the flow of fluid and the degree of atomization from a fine spray or mist to a coarse spray of oil. The nozzle shall be brass, and shall be designed in such manner as to allow easy adjustment without the use of tools. The sides of the nozzle shall be perforated, or other safety devices installed, to prevent oil/air pressure injury to personnel.

Source: https://assist.dla.mil -- Downloaded: 202610-18T03:26Z
Check the source to verify that this is the current version before use.

3.9 Operation. The oil gun shall spray in varying degrees of atomization as described in 3.8 with oils within a viscosity range of SAE 5W to SAE 30 inclusive, at temperatures ranging from plus 80° F to minus 20° F. The oil gun shall operate throughout its spray range with no evidence of malfunction, leakage, distorted spray pattern, or damage when tested as specified in 4.5.2.1 and 4.5.2.2.

3.10 Endurance. The oil gun shall withstand, without damage, 10,000 full trigger strokes of the oil gun trigger at an air pressure of 50 psig when tested as specified in 4.5.2.3. After the 10,000 cycles, the oil gun air valve shall hold entrapped air pressure of 100 psig for a period of not less than one minute with not more than 5 psig pressure loss when tested as specified in 4.5.2.3.1.

3.11 Identification marking. The oil gun shall be marked in accordance with MIL-STD-130.

3.12 Workmanship. All parts, components, and assemblies of the oil gun shall be clean and free from bulges, warpage, dirt, rust, defective connections and loose, damaged, or missing parts. External surfaces shall be free from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

Source: https://assist.dla.mil -- Downloaded: 202610-18T03:26Z
Check the source to verify that this is the current version before use.

4.3 First article inspection.

4.3.1 Examination. The oil gun shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The oil gun shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.1.1 Inspection lot. A lot shall consist of all oil guns of the same material offered for delivery at one time.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Tests. When specified (see 6.2), samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 6.5 percent defective.

4.5 Inspection procedures.

4.5.1 Examination. The oil gun shall be examined as specified herein for the following defects:

Major

101. Component missing.
102. Air inlet not located or threaded as specified.
103. Barrel not as specified.
104. Inability of thread to mate properly.
105. Fluid container not as specified.
106. Workmanship not as specified.

Minor

201. External surfaces not finished as specified.
202. Identification marking missing, incomplete or illegible.

Source: https://assist.dla.mil -- Downloaded: 202610-18T03:26Z
Check the source to verify that this is the current version before use.

4.5.2 Tests.

4.5.2.1 Low temperature. The oil gun shall be filled to capacity with SAE 5W oil and the gun and oil stabilized at minus 20° F plus or minus 5° F. The oil gun shall then be connected to an air supply of not more than 6 cubic feet per minute (cfm) at 50 psig pressure, and operated throughout its spray range until empty. Nonconformance to 3.9 shall constitute failure of this test.

4.5.2.2 Normal temperature. The oil gun shall be filled and operated as specified in 4.5.2.1, except that oil shall be SAE 30 and the temperature shall be plus 80° F, plus or minus 10° F. Nonconformance to 3.9 shall constitute failure of this test.

4.5.2.3 Endurance. The oil gun shall be subjected to the oil gun trigger test as specified in 3.10 within a 40 hour time period. Nonconformance to 3.10 shall constitute failure of this test.

4.5.2.3.1 Leakage. The oil gun shall be tested for leakage by connecting an air-pressure gage to the air inlet at the oil gun handle and connecting a 20 foot length of 5/16-inch diameter hose between the pressure gage and the air pressure source. An air pressure of 100 psig shall be applied and entrapped from the oil gun to the air source shut-off valve for a period of not less than 1 minute. Nonconformance to 3.10 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

107.	Method of preservation not as specified for Level A.
108.	Boxes not waterproofed sealed as specified for Level A preservation.
109.	Gross weight of boxes exceeds 200 pounds for Level A packing.
110.	Strapping not zinc-coated for Level A packing.
111.	Gross weight of boxes exceed the weight limitation of the box for Level B packing.
112.	Boxes not sealed and reinforced with tape as specified for Level B packing.
113.	Marking missing, illegible, incorrect or incomplete.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, Level B, or Commercial as specified (see 6.2).

5.1.1 Level A. Each oil gun, and instructions when furnished, shall be preserved together in accordance with MIL-P-116, Method III. Containers shall conform to PPP-B-636, W6c. After closure, the boxes shall be waterproof sealed with tape as specified for slotted style boxes in accordance with the appendix to the box specification.

5.1.2 Level B. Each oil gun, and instructions when furnished, shall be preserved together in accordance with MIL-P-116, Method III. Containers shall conform to PPP-B-636, Type CF, Class domestic, Variety SW, Grade 125. Closure shall be in accordance with the appendix to the box specification.

5.1.3 Commercial. Each oil gun, and instructions when furnished, shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. The oil guns, preserved as specified in 5.1, shall be packed in quantities not to exceed 200 pounds gross weight, in close-fitting boxes conforming to PPP-B-601, Overseas type, style optional. Closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Level B. The oil guns, preserved as specified in 5.1, shall be packed in quantities not to exceed the weight limitation of the boxes in close-fitting boxes conforming to PPP-B-636, V3c. The boxes shall be waterproof sealed with tape as specified for slotted style boxes, and reinforced with bands of tape in accordance with the appendix to the box specification.

5.2.3 Commercial. The oil guns, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.

5.3.2 Commercial. Marking for Commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The oil gun is designed primarily for use in spray application of oils to automotive and other equipment. It can also be used in cleaning and degreasing operations and as an air duster. In addition to lubricating oils, it is capable of spraying kerosene and other cleaning solvents.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When a first produced article is required for inspection and approval, and number of oil guns required (see 3.2, 4.3, and 6.3).
(c) When quality conformance testing is required (see 4.4.3).
(d) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(e) When other than Finish B strapping is required (see 5.2.1).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The first article should be a first produced oil gun. The first article should consist of one or more units. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the first article.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each oil gun.

6.5 Information figure. Figure 1 shows a type of oil gun which has been found acceptable; however, the figure is included for illustration only and is not intended to preclude the furnishing of another oil gun which conforms to this specification.

6.6 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review activities:
Army - AR, MD
DLA - CS

User activities:
Navy - MC

Preparing activity:
Army - ME

Project 4930-0293

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.

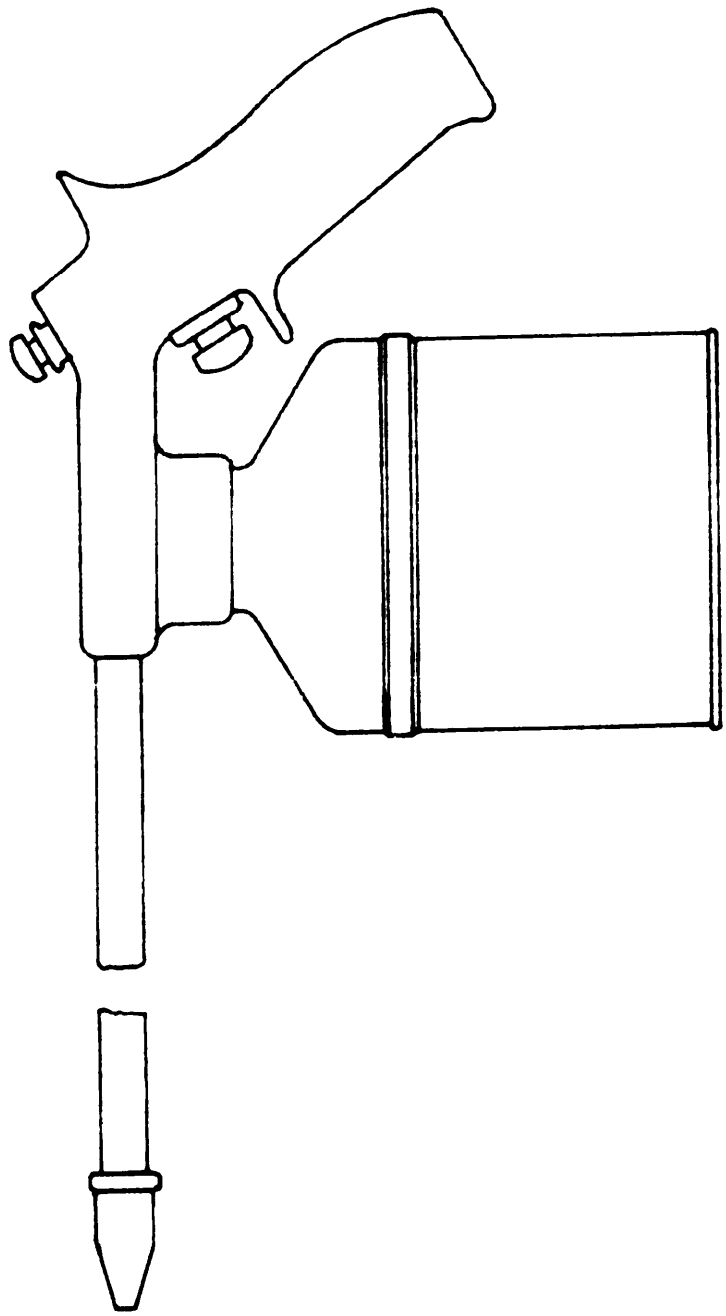


FIGURE 1. Oil gun pneumatic.

X-1647

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-O-14559C Oil Gun, Pneumatic

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.

LD

PARTMENT OF THE ARMY

OFFICIAL BUSINESS
ALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY CARD

FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Mobility Equipment Research and
Development Command
ATTN: DRDME-DS
Fort Belvoir, VA 22060

.D

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.